A Similarity-Based Ranking Method for Hyperspectral Band Selection

Buyun Xu, Xihai Li, Weijun Hou, Yiting Wang, and Yiwei Wei

***Abstract*— Band selection (BS) is a commonly used dimension reduction technique for hyperspectral images. In this article, we propose a similarity-based ranking (SR) strategy inspired by a density-based clustering algorithm. The representativeness of a band is evaluated according to its ability to become a cluster center. We introduce structural similarity (SSIM) to measure the relationships between the bands. Thus, our proposed ranking-based BS method is called SR-SSIM. We picked state-of-the-art BS methods as competitors and carried out classification experiments on different data sets. The results illustrated that SR-SSIM outperformed the other methods. It is demonstrated, in this article, that the SSIM is more suitable for hyperspectral BS than the Euclidean distance since the SSIM can mine the spatial information contained in the band images. Furthermore, we discuss the application of BS methods on deep learning classifier. We found that proper preprocessing by the BS method can effectively eliminate redundant information and avoid overfitting.**

***Index Terms*— Band selection (BS), hyperspectral images (HSIs), similarity measurement.**

I. Introduction

HYPERSPECTRAL sensors record the spectral responses of objects to many narrow electromagnetic bands with different wavelengths. A hyperspectral image (HSI) is produced by hyperspectral sensors and contains both spatial information and spectral information. HSI data form a 3-D data cube that can be regarded as a set of 2-D images and a set of 1-D spectral response curves. Compared with traditional visible and infrared remotely sensed images, HSIs have advantages in various applications because of their richer discriminative physical clues from the narrow continuous spectral bands. Currently, HSIs are attracting growing attention, and they have been widely used in both academia and industry, such as for biological analysis [1]–[3], medical imaging [4]–[6], and environmental monitoring [7]–[9].

However, the large dimensionality of HSI data causes a significant burden and a low efficiency of information processing [10], [11]. First, the large number of spectral bands corresponds to a large number of features and parameters when processing HSIs with machine learning methods, which results in complicated models and time-consuming processing [12]. Second, as more features are involved, more labeled training samples are needed, yet it is laborious and costly to obtain sufficient training samples in practice [13]. Third, there is considerable redundant information in HSI data due to the high correlation between neighboring bands, yet not all bands are equally important [14]–[16]. From these points of view, dimension reduction (DR) is a reasonable and necessary process for dealing with HSIs.

DR techniques can be achieved by feature extraction or selection (which is also called band selection (BS) for HSIs). The former approach projects the original data into a lower dimensional space, while the latter selects representative bands from the original band set. However, the extracted features of an HSI convey little physical meaning, while the selected features (or bands) can preserve the original spectral information, which is significant for geological analysis. Therefore, BS is more commonly used for HSIs.

In addition, based on the relationship with subsequent processes (such as classification and target detection), BS can be divided into filtered, wrapped, and embedded methods. Filtered methods regard BS as a preprocessing method of the application process. Wrapped methods adjust the band evaluation criterion according to the results of the HSI analysis, which often involves the participation of labeled samples. For embedded methods, the BS process is embodied in the application process. Here, we only concentrate on the filtered BS method, which can be seen as a process of data compression and information extraction, because this kind of BS method has wider applications.

Generally, BS aims to determine the bands that are most helpful for distinguishing ground objects and to eliminate redundant and invalid bands. To this end, the criterion to estimate the significance of bands and the strategy to search those qualified bands are the two key elements of BS. If there are labeled samples of the spectral characteristics involved in the BS process, i.e., in the supervised or semisupervised BS, the criteria to estimate bands usually aim to improve the class separability with the aid of training samples. However, for unsupervised BS, the criteria are more diversified, such as the variance and the Kullback–Leibler divergence of each band [17], correlation with other bands [18], [19], and description ability of a band [20]. In practice, the spectral characteristics of the same object may be divergent under different conditions. Since it is difficult to acquire labeled samples, unsupervised BS methods are more frequently used and applicable.

Just as BS criteria can be varied, there are many kinds of band search strategies. For instance, ranking-based methods

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quantify the importance of each band according to a predefined criterion and select top-ranked bands [13], [17]–[19]. Clustering-based methods first cluster all bands and then select the most representative band in each cluster to form the final band set [21], [22]. Iteration-based methods first generate a candidate subset and then repeatedly update the subset by discarding or adding one or several bands [23], [24]. In general, the search strategy often varies with the BS criterion.

In this article, we propose a ranking-based BS method via structural similarity measurement, which is called the SR-SSIM method. First, we introduce the structural similarity (SSIM) index, an index originally used for image quality assessment, to measure the similarity between each pair of bands. Meanwhile, we discuss how to choose the parameters of the SSIM index for HSIs. Inspired by the fast density-peak-based clustering (FDPC) [25] algorithm, we then propose a similarity-based ranking (SR) strategy that uses the priority to become cluster center as the criterion to assess the band representativeness. Our method can be regarded as an improved version of the enhanced FDPC (EFDPC) [13], which is an unsupervised BS method also inspired by FDPC. The main contributions are as follows.

1) By customizing the SSIM index for HSIs, we take the spatial information into account when measuring the relationship between different hyperspectral band images. To the best of our knowledge, this is the first time that SSIM has been used in the hyperspectral BS problem. In addition to the BS algorithm proposed in this article, the similarity metric can also be used for other hyperspectral BS methods.
2) We propose an SR strategy to evaluate the representativeness of the bands. By combining the average similarity and the significant dissimilarity of the bands, the SR can exploit more discriminative and informative bands simultaneously.
3) We show through experiments that, when deep learning is used for HSI classification, proper preprocessing by BS methods can effectively eliminate redundant information and avoid the overfitting phenomenon.

The rest of this article is organized as follows. In Section II, we review some state-of-the-art and up-to-date BS methods. In addition, the main ideas of the SSIM, FDPC, and EFDPC are briefly introduced. In Section III, we describe the framework of our proposed method in detail, which is divided into two parts: the similarity metric and the SR strategy. In Section IV, we verify the superiority of our method over several competitive methods through experiments on different HSI data sets and classifiers. Finally, we conclude our article in Section V.

II. Related Work

In this section, we first review some of the latest unsupervised BS methods and then briefly introduce the SSIM, FDPC, and EFDPC algorithms. The unsupervised BS methods can be divided into spectral-feature-based and spatial–spectral-feature-based methods.

A. Spectral-Feature-Based BS

Some BS methods based on spectral features view each band of HSI data as discrete points in the Euclidean domain. For example, the maximum ellipsoid volume (MEV) [26] is a BS method originally designed for multispectral BS problems, which aims to select three bands out of seven bands. It is based on the assumption that the bands with higher information and lower correlation can construct an ellipsoid with a larger volume, which means that the MEV can find informative and distinctive bands. With the development of hyperspectral sensors, the MEV has been extended for hyperspectral BS. For multispectral data, it is feasible to execute an exhaustive search among a total of $C_7^3 = 35$ combinations. However, for HSI data with hundreds of bands, it is impractical to use exhaustive strategies. Thus, the widely used suboptimal search methods, such as sequential backward selection (SBS) and sequential forward selection (SFS) [27], are often combined with MEV, forming the MEV-SBS [23] and MEV-SFS [24] methods, respectively. However, in each lookup round of SBS or SFS, the repetitive calculation of the volume is computationally complex and time-consuming.

Fortunately, it is found in volume-gradient-based band selection (VGBS) [23] that, when one band is removed tentatively, the volume of the remaining bands is proportional to the gradient of the whole simplex volume. In other words, the volume of the parallelotope is most sensitive to the most redundant band, which corresponds to the maximum volume gradient. Therefore, the repetitive calculation of the volume is unnecessary, and the computational complexity is greatly reduced. Thus, VGBS is a fast and equivalent version of the MEV-SBS method. In addition, a fast version of VGBS (FastVGBS) [28] was proposed later, which is mathematically equivalent to VGBS and can further accelerate the calculation process.

Moreover, the orthogonal-projection-based band selection (OPBS) [24] is a fast version of MEV-SFS. OPBS finds a subtle relationship between the orthogonal projections (OPs) of the candidate bands and the ellipsoid volume of the band set. Furthermore, the recursive formulas of the OPs are derived, and the incremental calculation of the OPs can be directly achieved.

These successive BS methods are all based on the assumption that the volume of the parallelotope constructed by the points can describe the information contained in the bands and that the Euclidean distance can reflect the relationship between bands. However, these methods are all sensitive to noise bands. The assumption that all bands are equivalent to points in a high-dimensional Euclidean space only mines the shallow spectral information in the HSI data, while the spatial information and contextual information are neglected [29]. This problem also exists in other BS algorithms that use the Euclidean distance to measure the relationship between bands, such as the EFDPC [13], adaptive distance-based band hierarchy (ADBH) [30], and adaptive subspace partition strategy (ASPS) [31] algorithms.

Deep learning methods have achieved great success in the computer vision field, and they have been increasingly applied

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to HSI processing in recent years. Furthermore, the attention mechanism in deep neural networks is often used in BS problems for HSIs. Specifically, the channel attention module aims to learn the relationship between one band and the other bands in the HSI data. Wang *et al.* [32] proposed a band attention network for HSI classification. When working in the BS mode, the network automatically learns to select the significant bands and discard the useless bands so that the classification accuracy can be maximized. Lorenzo *et al.* [33] coupled an attention-based convolutional neural network (CNN) with an anomaly detection technique for hyperspectral BS. The proposed method is an embedded approach that embeds the generation of attention heatmaps into CNN training. First, the attention heatmaps quantify the importance of specific parts of the spectrum, after which they are processed using an anomaly detection algorithm. Cai *et al.* [34] proposed a unified BS framework consisting of a band attention module (BAM) and a reconstruction network (RecNet). The BAM aimed to explicitly model the nonlinear interdependences between spectral bands, and the RecNet is used to restore the original HSI from the learned informative bands.

B. Spatial–Spectral-Feature-Based BS

BS methods based on spatial–spectral features make comprehensive use of spatial information and spectral information of HSI data. For instance, dual-clustering-based BS by context analysis (DCCA) [22] first defines a pairwise hyperspectral angle (PHA) descriptor containing the context information of both the neighboring bands and pixels. A dual-clustering algorithm was then developed from k-means clustering to jointly integrate the PHA feature and the raw HSI features. Furthermore, through a groupwise strategy for representation, the interactions between chosen bands were taken into account during the BS process.

In the BS method based on saliency bands and scale selection (SBSS) [35], the intrinsic connection between the saliency, scale, and image description is used to excavate the intrinsic properties of the HSI data. This method extends the concept of saliency and scale characteristics in 2-D image data to the 3-D HSI data. By defining a saliency index for the HSIs, the spatial information and spectral information are combined in the BS process.

The fast neighborhood grouping method for hyperspectral BS (FNGBS) [29] is a grouping-ranking method. First, based on the assumption that bands in the same group should have a continuous spectrum, the context information in a large spectral range is mined by partitioning the HSI data cube into several groups using a coarse–fine strategy. The bands in each group are then ranked according to the product of the local spectral density and the image spatial information entropy, which can obtain the most relevant and informative bands simultaneously.

The BS method called the efficient graph convolutional self-representation (EGCSR) method [36] incorporates graph convolution into the self-representation model. The method makes full use of the structural information and tends to result in a robust self-representation coefficient matrix because it is modeled in the non-Euclidean domain.

Roy *et al.* [37] proposed a dual-attention reconstruction network for BS (DARecNet-BS). The network is an end-to-end network, and a position attention module (PAM) and a channel attention module (CAM) are employed simultaneously to recalibrate the feature maps. Thus, this method can capture the long-range nonlinear contextual information in both spatial and spectral directions.

C. Structural Similarity

The SSIM index was original proposed for image quality assessment to compare a noisy image with a raw image [38]. Different from the Minkowski error metric, which is based on pointwise signal differences and is independent of the underlying signal structure, structural similarity measurement compares the structures of the reference and the distorted signals. The system separates the task of similarity measurement into three comparisons: luminance, contrast, and structure, which is expressed as follows:

$$\mathrm{SSIM}(x, y) = l(x, y) \cdot c(x, y) \cdot s(x, y). \tag{1}$$

Supposing that $x \in \mathbb{R}^{M\times N}$ and $y \in \mathbb{R}^{M\times N}$ are two image signals, the luminance of signal x is estimated as the mean intensity $\mu_x = (1/(M \times N)) \sum_{p=1}^{M\times N} x_p$, where x_p denotes the gray value of a pixel. The contrast of the signal is unbiasedly estimated as the standard deviation (square root of variance)

$$\sigma_x = \sqrt{\frac{1}{M \times N - 1} \sum_{p=1}^{M\times N} \left(x_p - \mu_x\right)^2}.$$

The luminance comparison function is defined as follows:

$$l(x, y) = \frac{2\mu_x\mu_y + C_1}{\mu_x + \mu_y + C_1} \tag{2}$$

where C_1 is a constant that is included to avoid instability when $\mu_x + \mu_y$ is close to 0. The constants C_2 and C_3 are included for the same reason. Similarly, the contrast comparison function takes the form

$$c(x, y) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x + \sigma_y + C_2}. \tag{3}$$

The structure comparison function is the cosine similarity of two standardized image signals

$$s(x, y) \triangleq \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3} \tag{4}$$

where

$$\sigma_{xy} = \frac{1}{M \times N - 1} \sum_{p=1}^{M\times N} \left(x_p - \mu_x\right)\left(y_p - \mu_y\right).$$

We set $C_3 = C_2/2$ to simplify the expression. Thus, the resulting form of the SSIM index is as follows:

$$\mathrm{SSIM}(x, y) = \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\sigma_{xy} + C_2\right)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)}. \tag{5}$$

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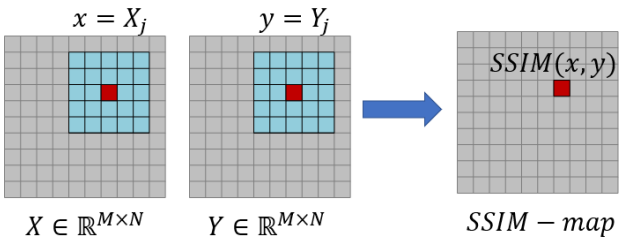


Fig. 1. Local SSIM in one window.

Specifically, C_1 and C_2 were chosen as $(0.01\times L)^2$ and $(0.03\times L)^2$, respectively, in a previous publication [38], where L is the dynamic range of the pixel values.

Since image statistical features are usually highly spatially unstable, the SSIM index is more useful when it is applied locally rather than globally. As previously reported [38], the local statistics and SSIM index are calculated within a local 11×11 square window, and an 11×11 circular-symmetric Gaussian weighting function $\Omega = \{\omega_i \mid i = 1, 2, \ldots, (11 \times 11)\}$, which has a standard deviation of 1.5 samples and is normalized to a unit sum $(\sum_{i=1}^{11\times 11} \omega_i = 1)$, is employed. The estimates of the local statistics are then modified accordingly as follows: $\mu_{x_\text{local}} = \sum_{i=1}^{11\times 11} \omega_i x_i$, $\sigma_{x_\text{local}} = (\sum_{i=1}^{11\times 11} \omega_i (x_i - \mu_x)^2)^{1/2}$, and $\sigma_{xy_\text{local}} = \sum_{i=1}^{11\times 11} \omega_i (x_i - \mu_x)(y_i - \mu_y)$.

As shown in Fig. 1, the window moves pixel-by-pixel over the entire image, and a mean SSIM (MSSIM) index is obtained to evaluate the overall similarity of two images

$$\text{MSSIM}(X, Y) = \frac{1}{J}\sum_{j=1}^{J} \text{SSIM}\left(X_j, Y_j\right). \tag{6}$$

D. Fast Density-Peak Clustering

In an article published in *Science* in June 2014, an FDPC algorithm was proposed [25]. The core idea of the FDPC lies in the description of cluster centers. This algorithm identifies cluster centers that have two characteristics:

1) a higher local density, denoted by ρ,
2) a larger distance from the nearest point with a higher density, denoted by δ.

Based on these two factors, the importance of all the points is qualified and ranked. The top-ranked points are then chosen as cluster centers. The idea of the FDPC algorithm is a highly suitable approach for the BS problem since it can quickly identify the cluster centers. Jia improved this clustering algorithm and applied it to the BS process, which is called the EFDPC [13]. The improvements of the EFDPC over the FDPC algorithm are given as follows.

1) It uses an exponential-based heuristic rule for the cutoff threshold d_c.
2) ρ and δ are normalized to the scale of $[0, 1]$, and the score $\gamma = \rho \times \delta^2$.

These two improvements of the EFDPC are reasonable because the BS problem aims at selecting a certain number of bands, which can be performed by determining a given number of cluster centers. Thus, as k increases (i.e., the number of cluster centers increases), data points should be divided into smaller clusters. Therefore, the cutoff distance d_c is regularized by an exponential-based heuristic rule related to k, and d_c gradually decreases with the increase in k. In addition, because the range of ρ is much larger than that of δ, normalization can eliminate the dimension difference between δ and ρ. The square operation on δ can compensate for ρ for the loss of weight during the normalization procedure. However, balancing the weights of the density ρ and the distance δ requires great care. If ρ is given more weight than δ, the selected band subset may contain more redundant information; otherwise, the selected band subset may be more vulnerable to noise pollution. However, in the EFDPC, through the square operation on δ, the values of the final relative distance δ^2 are more compact for the closer points but sparser for the farther points, as shown in Fig. 2. This approach fails to distinguish small clusters in dense areas but increases the influence of isolated noise points.

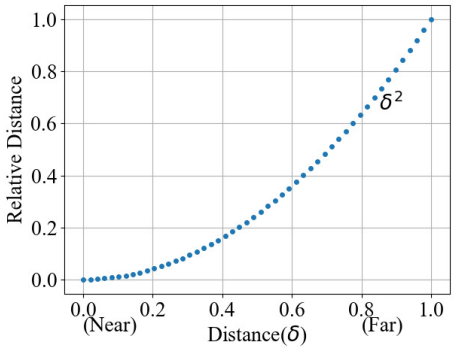


Fig. 2. Square operation on the distance δ.

III. Methodology

In this section, our proposed SR-SSIM framework will be presented in detail. We first introduce the approach to construct the similarity matrix and then define the SR strategy. For ease of description, we let one 3-D matrix $H \in \mathbb{R}^{M\times N\times L}$ denote the raw data cube of the HSI. This means that the HSI contains $M \times N$ pixels and L spectral bands. In addition, the number of selected bands is k.

A. Similarity Metric for Hyperspectral Band Images

1) Why Choose SSIM for Metric?: As is well known, when dealing with clustering problems, the distance or similarity measurement has a crucial and direct impact on the results. Compared with the general clustering process, the BS process aims at clustering images rather than some discrete points. In terms of the image similarity measurements, the mean square error (MSE), the peak signal-to-noise ratio (PSNR), and SSIM are commonly used indexes. Supposing that $x \in \mathbb{R}^{M\times N}$ and $y \in \mathbb{R}^{M\times N}$ are two image signals,

$$\text{MSE} = \frac{1}{M \times N}\sum_{i=1}^{M}\sum_{j=1}^{N}\left(x_{ij} - y_{ij}\right)^2 \tag{7}$$

$$\text{PSNR} = 10\log_{10}\frac{\left(2^b - 1\right)^2}{\text{MSE}} \tag{8}$$

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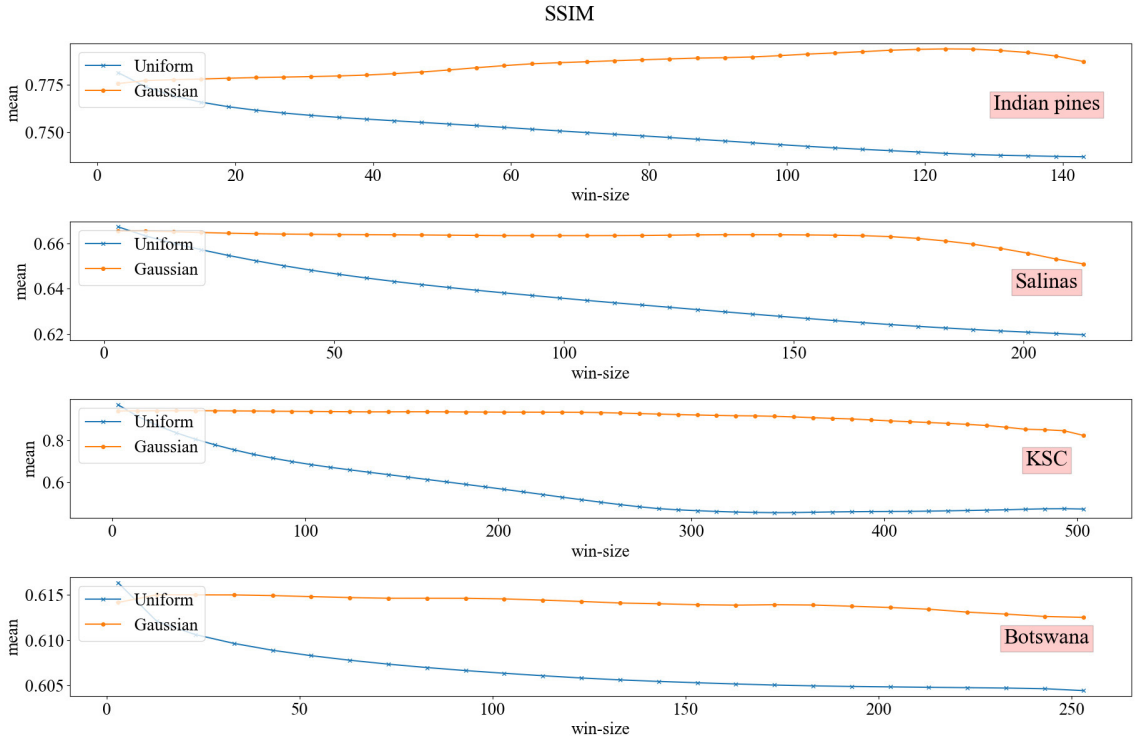


Fig. 3. Mean value of similarity matrix versus window size and kernel type.

where b is the number of bits per pixel. The MSE is equal to the Euclidean distance, while the PSNR was defined based on the MSE. In the Euclidean domain, the BS process for HSIs is based on a model that selects k points from L points in a $(M \times N)$-dimensional space, while $L \ll (M \times N)$. Thus, the model is equivalent to processing sparse data points in a super-high-dimensional space [39]. In general, the similarity metrics based on distances are sensitive to variations within a data distribution or the dimensionality of a data space [40]–[43]. For L_k norms in high dimensions

$$\lim_{d\to\infty} \operatorname{var}\left(\frac{\|X_d\|_k}{E(\|X_d\|_k)}\right) = 0, \quad \text{then } \frac{D\max_d^k - D\min_d^k}{\operatorname{Dmin}_d^k} \to_p 0 \tag{9}$$

which means that, under certain reasonable assumptions about the data distribution, the difference between nearest and farthest neighbors to a given target will diminish as the dimensionality increases. The behavior of the integer L_k norm in high-dimensional spaces reduces the utility of the metrics for discrimination. Therefore, the Euclidean distance, based on our intuition from a 3-D world, is not a good metric for sparse data in a high-dimensional space.

Since SSIM performs well for the similarity measurement between natural images, it is widely applied as an image similarity measure. In fact, HSIs are natural image signals that are highly structured [22]. Furthermore, the pixels of HSIs show strong dependencies, especially when they are spatially adjacent, and these dependencies carry important information about objects in a visual scene [35], [38]. In the study of HSIs, the SSIM is usually used to evaluate the result of super-resolution reconstruction or denoising [44], [45]. In this article, the SSIM is introduced to evaluate the band relationship for hyperspectral BS.

2) Customizing SSIM for Hyperspectral Band Images: For an HSI denoted by a 3-D matrix $H \in \mathbb{R}^{M\times N\times L}$, we can construct the structural similarity matrix $S \in \mathbb{R}^{L\times L}$ by calculating the structural similarity between band image pairs. Letting $H_l \in \mathbb{R}^{M\times N}$ denote the lth band image, S_{ij} indicates the similarity between the ith and jth band images

$$S_{ij} = \operatorname{MSSIM}(H_i, H_j). \tag{10}$$

Compared with the natural image, there is no spatial distortion between various band images of the HSI, and the difference between them is mainly due to the inconsistency of different ground objects in various spatial positions with spectral changes. In addition, the spatial resolution of the HSI is usually low. Therefore, when using the SSIM index algorithm to measure the relationship between different band images of HSIs, we should consider how to adjust the parameters of the SSIM to properly reflect the spatial structure information based on the characteristics of the HSIs.

It has been found in experiments that the performance of the SSIM index algorithm is not sensitive to the values of K_1 and K_2 [38]. Thus, in this article, we examine the impact of the window size on the similarity matrix of the HSI. Figs. 3 and 4 show the curves for four different data sets, where the mean value and standard deviation of the similarity matrix (S) vary with the window size. In addition, the difference between the uniform kernel and the Gaussian kernel is represented by two different colors. The statistics of S vary gently with the window size when adopting a

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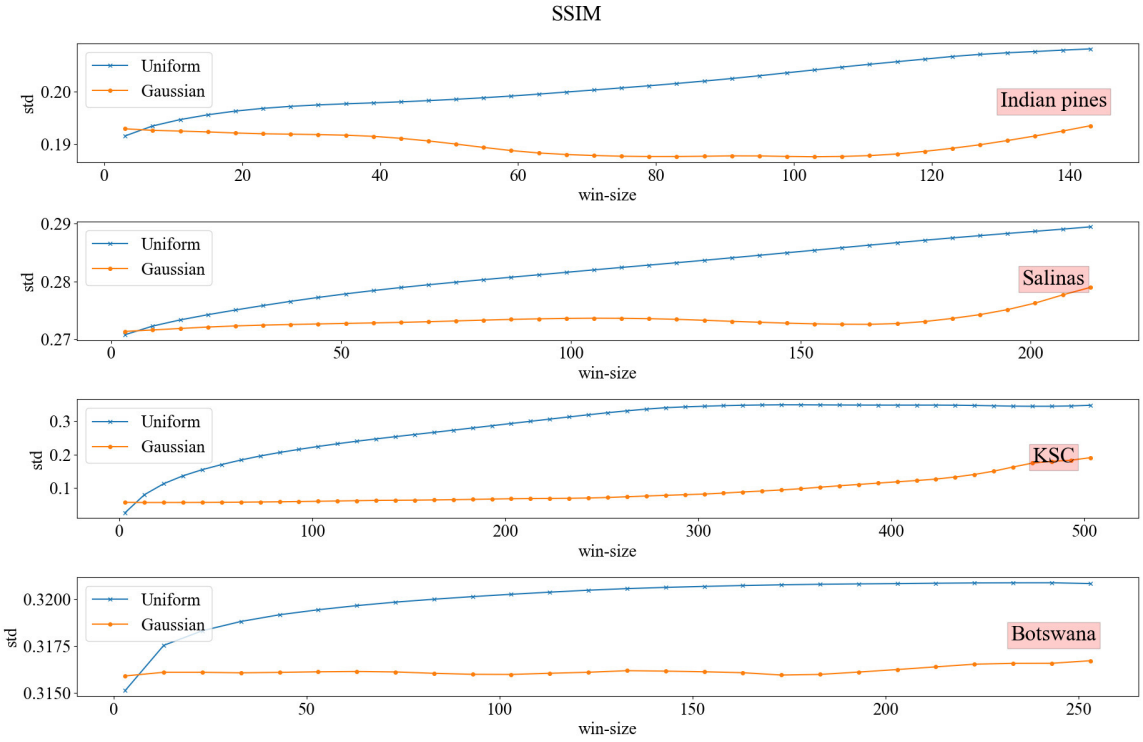


Fig. 4. Standard deviation of similarity matrix versus window size and kernel type.

Gaussian kernel. In addition, assuming the window size is w, the complexity of calculating the similarity of S is $O(w) = (M-w+1)(N-w+1)w^2L^2$, which varies with the value of w, as shown in Fig. 5. Therefore, to reduce the computational complexity and fully mine the details of the image spatial information, we selected $w = 3$ for the SSIM index algorithm to construct the similarity matrix of the HSI. The notation "SSIM-w3" is used to represent the setting with $w = 3$, and "SSIM" represents the original parameters presented in a previous publication [38].

B. Similarity-Based Ranking

We propose an SR strategy inspired by the FDPC. Instead of using the two key factors (density ρ and distance δ) in the FDPC, we define the average similarity α and significant dissimilarity θ for each band based on the similarity matrix, which is more adaptable for hyperspectral BS problems.

1) Average Similarity: The average similarity α of band i is defined as

$$\alpha_i = \begin{cases} \text{mean}\left(S_{ij}\right), & S_{ij} > s_c \\ 0, & S_{ij} \le s_c \end{cases} \tag{11}$$

where s_c is a cutoff similarity threshold used to exclude those bands with low similarity and weak correlation. We sort the values of the similarity lower triangular matrix from low to high as follows:

$$\{s_1 \le s_2 \le \cdots \le s_Y\} = \text{tril}(S). \tag{12}$$

We then choose s_c as

$$s_c = \text{mean}(s_m + s_{m+1} + \cdots + s_n) \tag{13}$$

where $Y = L(L-1)/2, m = \text{round}(Y \times 5\%)$, and $n = \text{round}(Y \times 10\%)$, that is, s_c is the mean value from the mth position to the nth position in tril(S). In this way, s_c can adapt to different data based on the data distribution. Compared with the cutoff distance d_c in EFDPC, the value of s_c is flexible. In addition, the value of α is independent of the number of selected bands (k) and is only related to the HSI data itself. Therefore, it can reduce the computational cost, while the size of the band subset must be changed in practice.

2) Significant Dissimilarity: The significant dissimilarity θ of band i is defined as

$$\theta_i = \sqrt{1-\varphi_i^2} \tag{14}$$

where φ_i is the similarity between band i and the nearest band with a larger average similarity α, i.e.,

$$\varphi_i = \begin{cases} \max_{j:\alpha_j > \alpha_i}\left(S_{ij}\right), & \text{if } \alpha_i \ne \max(\alpha) \\ \min_{j \in I_s \setminus \{i\}}\left(\varphi_j\right), & \text{if } \alpha_i = \max(\alpha). \end{cases} \tag{15}$$

The unit arc function is adopted to map φ to θ, which is shown in Fig. 6 Hence, the value of θ_i can indicate the dissimilarity between band i and the nearest band with a larger average similarity. On the one hand, it enlarges the dissimilarity between near bands, which helps to select more representative bands in relatively dense groups. On the other hand, it smooths the dissimilarity between far bands, which contributes to reducing the impact of outlying noise bands.

3) Score of Each Band: The representativeness of each band can be evaluated by combining the average similarity (α) and significant dissimilarity (θ). As the values of the two factors do not have the same order of magnitude, they are normalized

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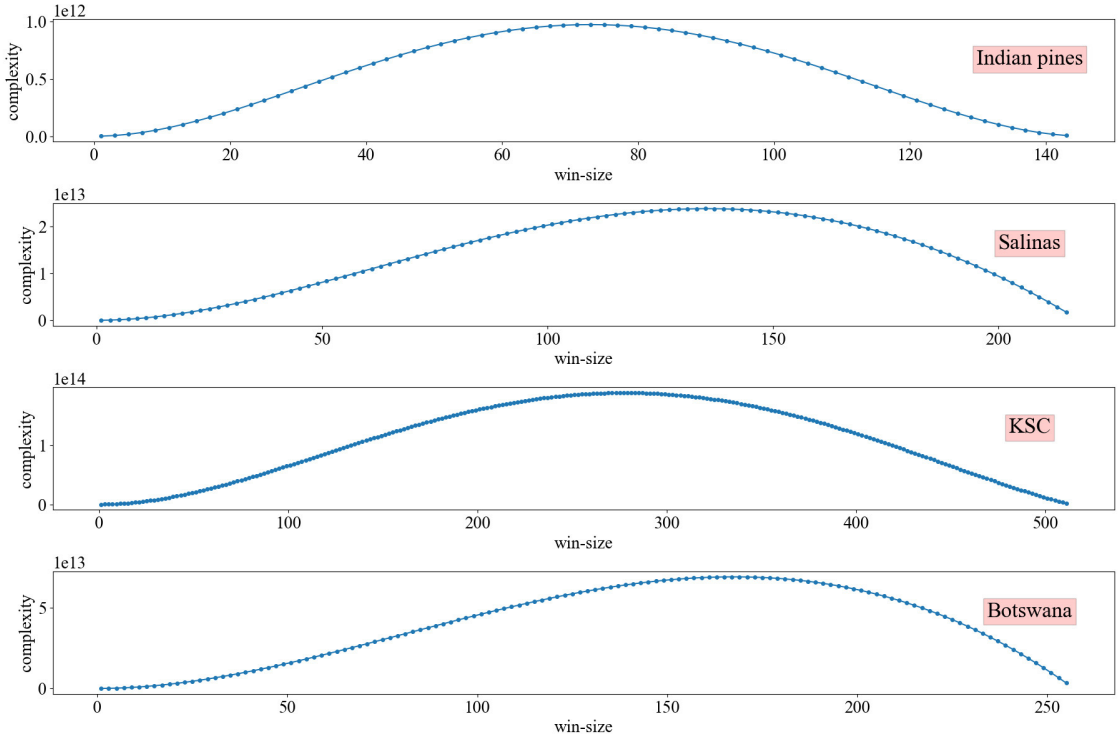


Fig. 5. Computational complexity of similarity matrix for HSI versus window size.

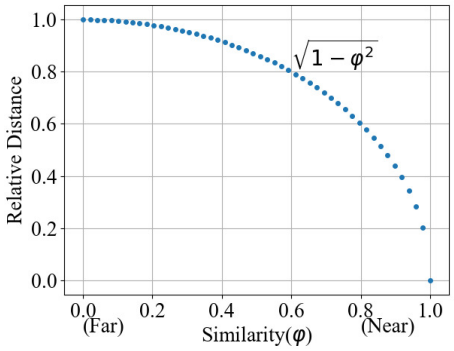


Fig. 6. Operation on similarity φ.

first, as follows:

$$\begin{aligned} \text{norm}(\alpha_i) &= (\alpha_i - \alpha_{\min})/(\alpha_{\max} - \alpha_{\min}) \\ \text{norm}(\theta_i) &= (\theta_i - \theta_{\min})/(\theta_{\max} - \theta_{\min}). \end{aligned} \tag{16}$$

Finally, all bands are ranked according to the product of the average similarity and the significant dissimilarity, and the first k bands are selected to form the band subset, as follows:

$$\eta_i = \text{norm}(\alpha_i) \times \text{norm}(\theta_i). \tag{17}$$

The pseudocode of SR is displayed in Algorithm 1, and the flowchart of the SR-SSIM is illustrated Fig. 7.

IV. Experiments

The efficacy of hyperspectral BS is often mediately evaluated by extensive experiments, such as HSI classification. In this section, we assess the performance of our proposed method from two aspects.

1) *Application on Conventional Classifiers:* Pixel-oriented classification is implemented as the follow-up application of BS. Higher classification accuracy means that the selected bands are more discriminative and the DR of the HSI is more efficient.
2) *Application on Deep Learning Classifier:* Deep learning models have shown promising results for the classification task. We examine whether adding BS as a preprocessing step can improve the accuracy of the deep neural network.

A. Competitors and Data Sets

1) Competitors: To verify the performance of the proposed method SR-SSIM, we compared it to several BS methods.

1) EFDPC [13], which was shown to be superior to ID [18], MVPCA [17], and AP [46]. EFDPC is the target algorithm we want to improve.
2) OPBS [24], which was demonstrated to be superior to LCMV-BCC [18], LCMV-BCM [18], MVPCA, MI [47], VGBS [23], and MR [48]. As described in Section II-A, OPBS is a typical geometry-based approach in the Euclidean domain.
3) TRC-OC-FDPC and NC-OC-FDPC [21], which were proven to be better than WaLuDi [49], UBS [18], VGBS, EFDPC, and MTSP [20]. TRC-OC-FDPC and NC-OC-FDPC are two forms of optimal clustering framework (OCF) proposed by Wang *et al.* [21]. The OCF combines band clustering and ranking strategies by

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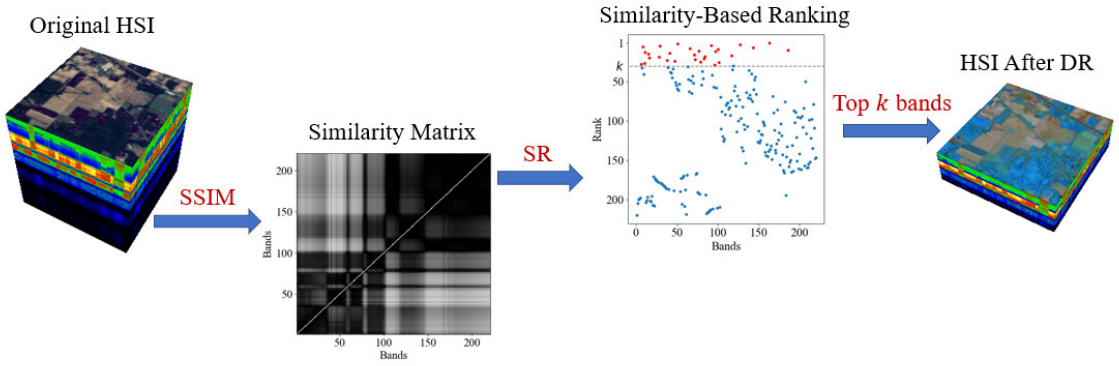


Fig. 7. Flowchart of SR-SSIM to conduct BS. First, we calculate the structural similarity between bands. We then obtain the score of each band based on the average similarity and the significant similarity. Finally, we rank all the bands according to the representativeness score and select the top k bands.

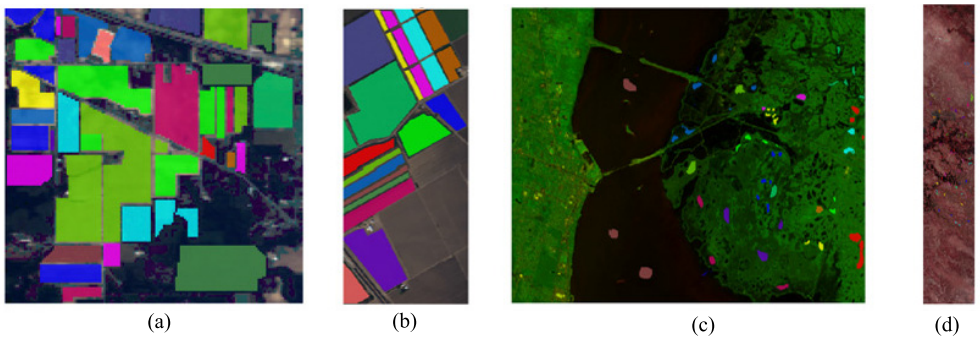


Fig. 8. Ground truth of these data sets in our experiment. (a) Indian Pines. (b) Salinas. (c) KSC. (d) Botswana.

selecting the top-ranked band in each group individually. TRC and NC are two objective functions, which are used in TRC-OC-FDPC and NC-OC-FDPC, respectively, and EFDPC is a sort of ranking criterion.

4) EFDPC-SSIM, which is formed by replacing the Euclidean distance in the EFDPC algorithm with (1 − SSIM). The EFDPC-SSIM method is used for the ablation study.

2) HSI Data Sets: Since we used pixel-oriented classification as the application test of BS, four benchmark data sets were chosen. They are listed in Table I and displayed in Fig. 8 with overlayed class masks.

The Indian Pines data set was acquired by an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over agricultural land in northwestern Indiana's Indian pine test site in 1992. The most commonly used image file consists of 145 × 145 pixels and 224 spectral bands. The spatial resolution is 20 m/pixel, and the spectral resolution is about 10 nm, while the spectral wavelength ranges from 400 to 2500 nm. After discarding four zero bands, 20 bad bands disturbed by water absorption, and 15 severe noisy bands with low signal-to-noise ratios (SNRs), the Indian Pines data set used in this study had 185 bands.

The Salinas scene data set was gathered by AVIRIS in 1998. Similar to the Indian Pines data set, the Salinas Scene data contain 224 bands with 10-nm widths and wavelengths ranging from 400 to 2500 nm. After discarding 20 bands of water absorption and low SNR, 204 bands were retained. The data contained 512 × 217 pixels but had a spatial resolution of 3.7 m/pixel. For classification purposes, 16 classes of ground objects were defined.

The Kennedy Space Center (KSC) data set was also collected by AVIRIS over the Kennedy Space Center, FL, USA, on March 23, 1996. It has a spatial resolution of 18 m/pixel and consists of 224 bands. In the provided file, 48 water absorption and low SNR bands were discarded, resulting in a total of 176 bands. The image file was selected using land-cover maps derived from color infrared photography provided by the KSC and Landsat Thematic Mapper (TM) imagery. A vegetation classification scheme was developed by KSC personnel in an effort to define functional types that are discernable at the spatial resolution of the Landsat and these AVIRIS data. There were 512 × 614 pixels and 13 classes in this image.

The Botswana data set was acquired by the NASA EO-1 satellite over the Okavango Delta, Botswana on May 31, 2001. The Hyperion sensor on EO-1 acquired data at a 30-m/pixel resolution in 242 bands covering the 400- to 2500-nm portion of the spectrum in 10-nm windows. The UT Center for Space Research preprocesses the data by mitigating the effects of bad detectors, interdetector miscalibration, and intermittent anomalies. The uncalibrated and noisy bands that covered the water absorption features were removed, and thus, only 145 bands were provided for study. The image includes 14 identified classes.

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Algorithm 1 SR Algorithm

Input: $S \in \mathbb{R}^{L\times L}$: the similarity matrix; s_c: the cutoff similarity threshold; K: the desired number of selected bands.

Output: M: the indexes of the cluster centers.

Begin
/* **Step1: the Average Similarity of Each Band** (α) */
$\alpha_i = 0, \quad (i = 1, 2, \ldots, L)$
$t = 0$
for $i = 1$ *to* L **do**
  for $j = 1$ *to* L **do**
    if $S_{ij} > s_c$ **then**
      $\alpha_i = \alpha_i + S_{ij}$
      $t = t + 1$
    end
  end
  $\alpha_i = \alpha_i / t$
end
$[\xi, I] = \text{sort}(\alpha, \text{'descend'})$
/* **Step2: the Significant Dissimilarity of Each Band** (θ) */
$\varphi_{I_1} = 1; \varphi_{I_i} = 0 \quad (i = 2, 3, \ldots, L)$
for $i = 2$ *to* L **do**
  for $j = 1$ *to* $i - 1$ **do**
    if $\varphi_{I_i} < S_{I_i I_j}$ **then**
      $\varphi_{I_i} = S_{I_i I_j}$ % φ: the similarity to the most similar band with higher average similarity.
      $A_{I_i} = I_j$ % A: the index of the most similar band with higher average similarity.
    end
  end
end
$\varphi_{I_1} = \min(\varphi)$
$A_{I_1} = I_1$
$\theta = \sqrt{1 - \varphi^2}$
/* **Step3: the Score of Each Band** (η) */
$\text{norm}(\alpha) = (\alpha - \alpha_{\min})/(\alpha_{\max} - \alpha_{\min})$
$\text{norm}(\theta) = (\theta - \theta_{\min})/(\theta_{\max} - \theta_{\min})$
$\eta = \text{norm}(\alpha) \times \text{norm}(\theta)$
/* **Step4: the Indexes of Selected Bands** (M) */
$[\zeta, G] = \text{sort}(\eta, \text{'descend'})$
$M = G(1 : K)$
End

B. Application on Conventional Classifiers

1) Experiment Settings:

The settings for the application experiments on conventional classifiers are listed below.

1) *Data Set Split:* To simplify the classification problem, the background of each data set was not considered. In the experiments, 10% of the samples for each class were chosen randomly to form the training set, while the remaining 90% were used for the test. Furthermore, to reduce the instability caused by the random split of the training and testing samples, the final classification results were obtained by averaging 20 individual runs.
2) *Classifiers:* To ensure objectivity in the evaluation process, we employed three classifiers: support vector machine (SVM), multinomial logistic regression (MLR), and random forest (RF). All classifiers were implemented in Python using scikit-learn, with the parameters set at the default values.
3) *Accuracy Assessment:* We used four indexes to assess the classification results: overall accuracy (OA), Cohen's kappa coefficient (Kappa), macroprecision, and macrorecall. Among them, macroprecision means the user's accuracy, i.e., 1 minus the commission error, and the macrorecall refers to the producer's accuracy, i.e., 1 minus the omission error.

TABLE I
INFORMATION OF THESE DATA SETS IN OUR EXPERIMENT

HSI names	Pixels	Spatial resolutions	Classes	Bands
Indian Pines	145 × 145	20 m/pixel	16	185
Salinas	512 × 217	3.7 m/pixel	16	204
KSC	512 × 614	18 m/pixel	13	176
Botswana	1476 × 256	30 m/pixel	14	145

2) Results and Analysis: The four indicators described above are presented below as functions of the number of selected bands for the interval (5, 10, …, 50).

For the Indian Pines data set, the classification results produced by the SVM, MLR, and RF classifiers are illustrated in Fig. 9. As depicted, SR-SSIM and SR-SSIM-w3 were completely superior to the other methods based on the SVM classifier. For the MLR and RF classifiers, SR-SSIM and SR-SSIM-w3 performed comparably to TRC-OC-FDPC and NC-OC-FDPC based on the OA and Kappa indexes, and they outperformed the other methods. In addition, the EFDPC-SSIM also achieved better performance than the EFDPC but did not perform SR-SSIM and SR-SSIM-w3.

The classification results for the Salinas scene are presented in Fig. 10. Similar to the Indian Pines scene, SR-SSIM and SR-SSIM-w3 achieved the best performances again. EFDPC-SSIM performed better than EFDPC but was not as stable as SR-SSIM and SR-SSIM-w3. Among the BS methods used in the experiments, TRC-OC-FDPC and OPBS performed most unsteadily with the different classifiers.

The results for the KSC data are shown in Fig. 11. SR-SSIM and SR-SSIM-w3 only needed ten bands to be selected to achieve very good results, while EFDPC and EFDPC-SSIM required more than 20 bands to be selected to achieve better performance. NC-OC-FDPC performed SR-SSIM and SR-SSIM-w3, and TRC-OC-FDPC achieved the best performance. In addition, OPBS performed the worst on this data set.

The OA curves for the Botswana data set are displayed in Fig. 12. SR-SSIM and SR-SSIM-w3 were superior to the other four methods, and they achieved satisfactory accuracy when only five bands were selected using the SVM classifier. Similar to the case for Indian Pines-185, NC-OC-FDPC also

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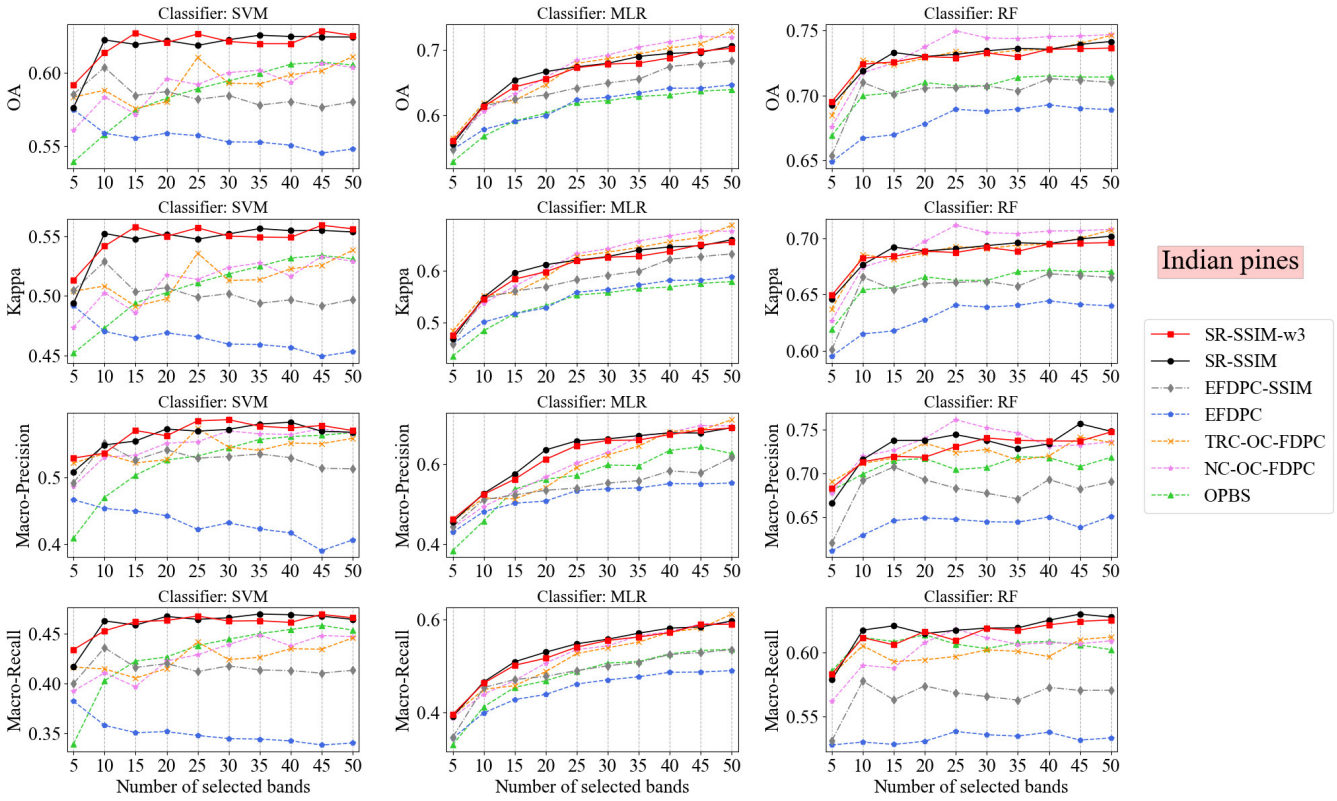


Fig. 9. OA, kappa, macroprecision, and macrorecall curves for the Indian Pines data set using the SVM, MLR, and RF classifiers with different BS methods.

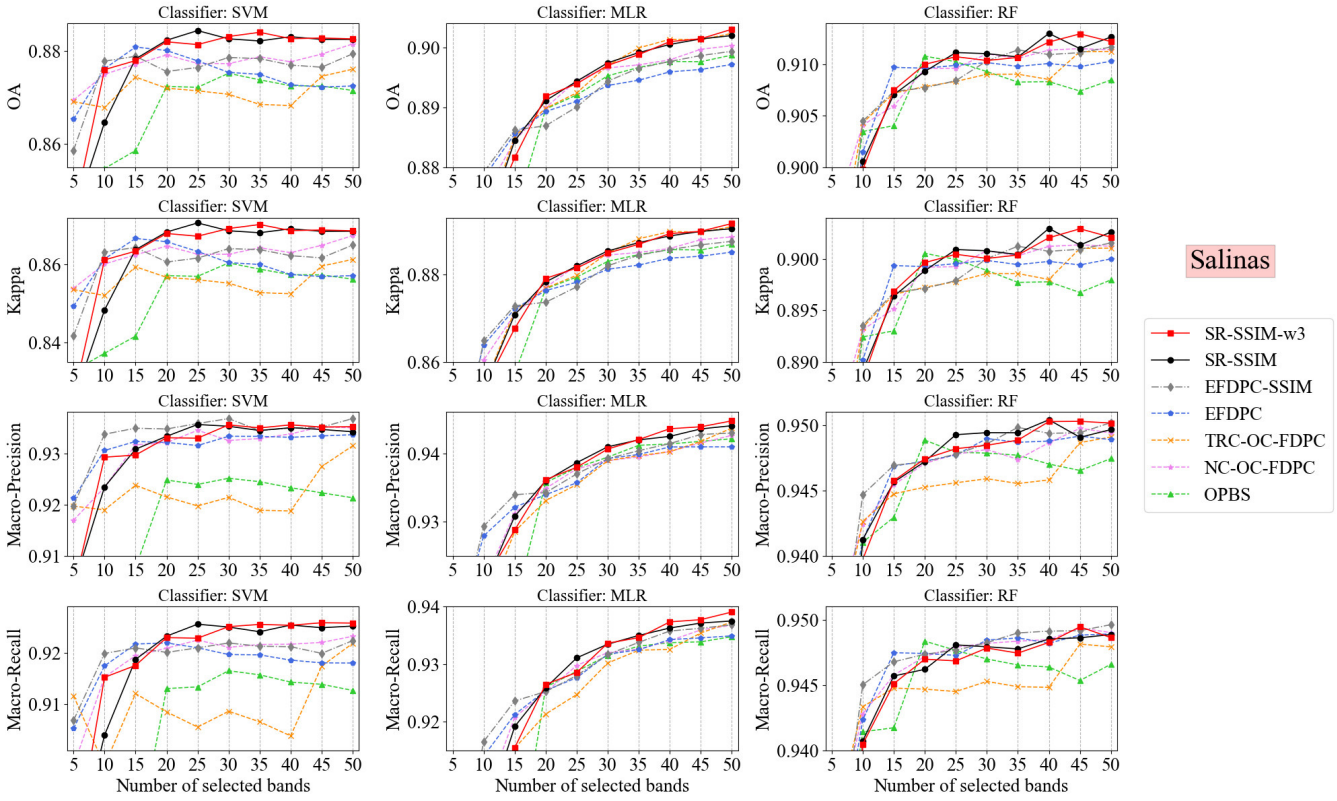


Fig. 10. OA, kappa, macroprecision, and macrorecall curves for the Salinas data set using the SVM, MLR, and RF classifiers with different BS methods.

performed well. The performance of EFDPC-SSIM was better than that of EFDPC but worse than those of SR-SSIM and SR-SSIM-w3.

C. Application on Deep Learning Classifier

1) Experiment Settings: In recent years, deep learning has become a popular research topic. In this section, we explore

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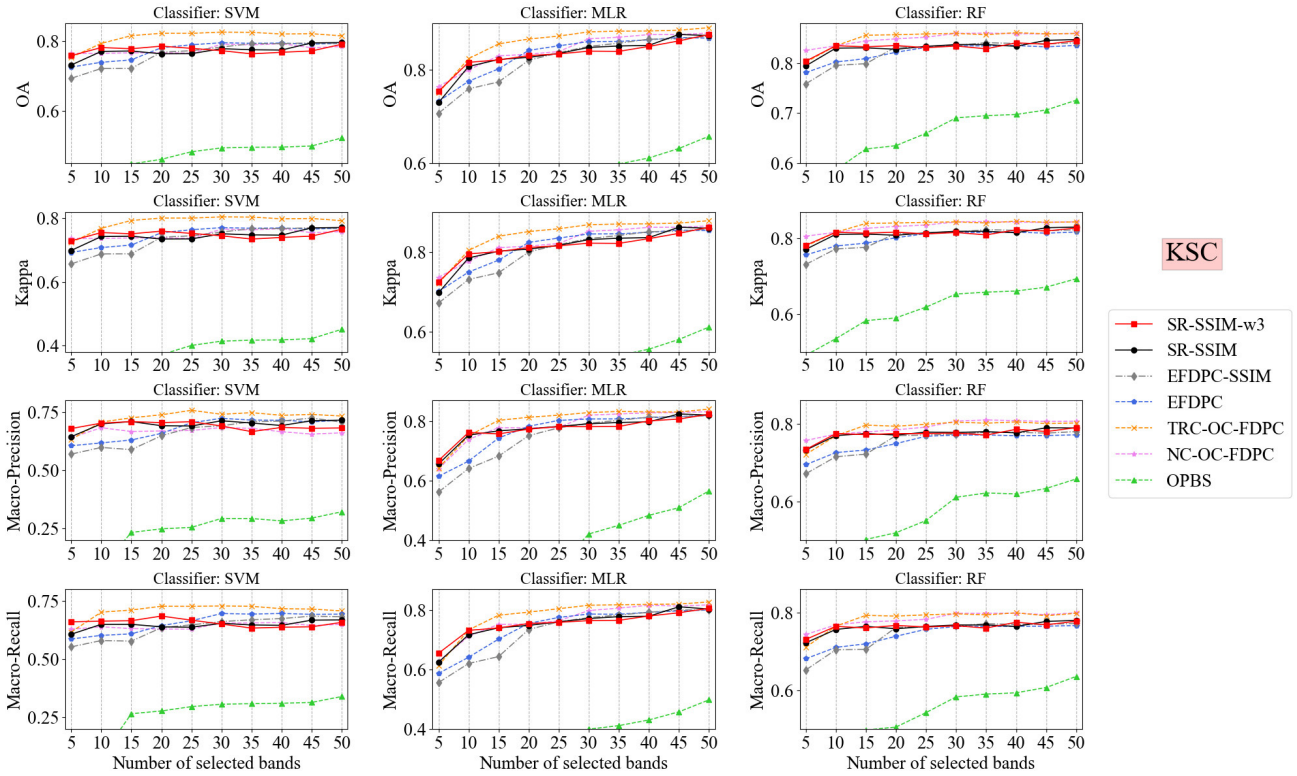


Fig. 11. OA, kappa, macroprecision, and macrorecall curves for the KSC data set using the SVM, MLR, and RF classifiers with different BS methods.

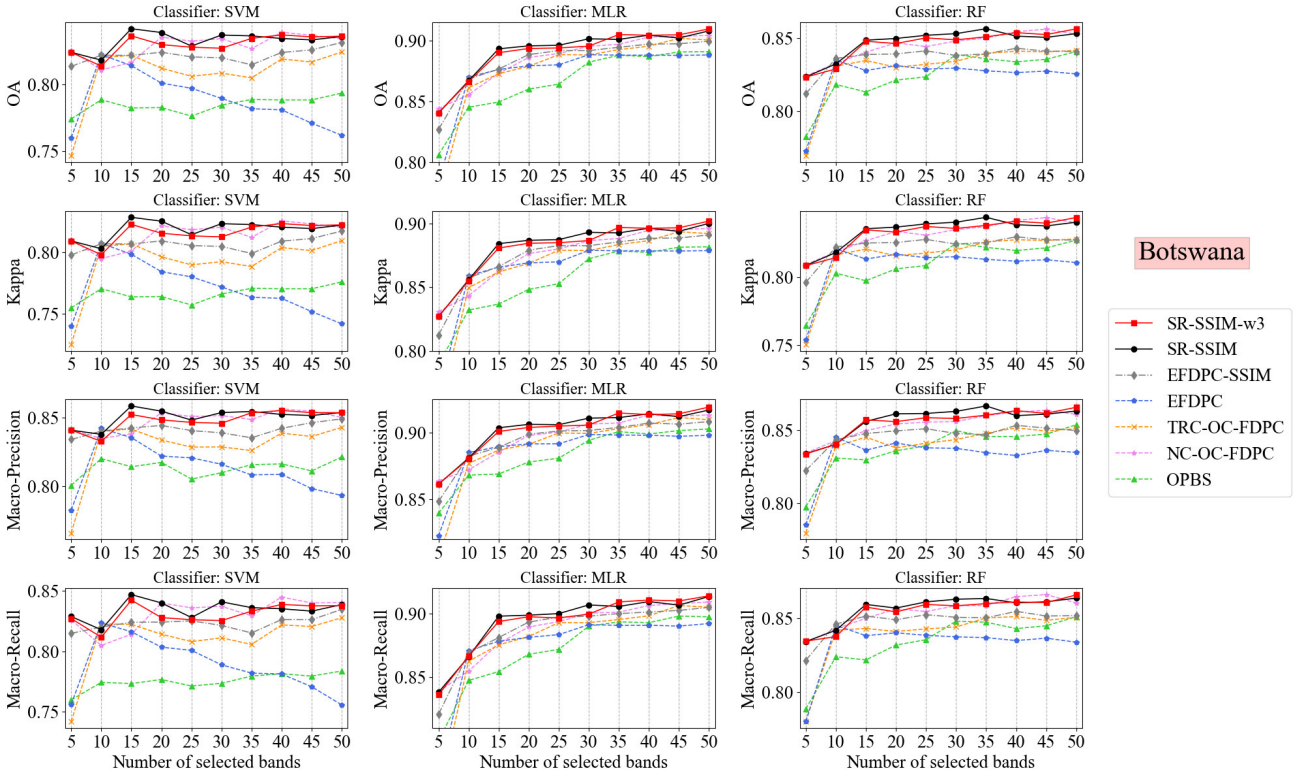


Fig. 12. OA, kappa, macroprecision, and macrorecall curves for the Botswana data set using the SVM, MLR, and RF classifiers with different BS methods.

the application of the BS method on the deep learning network in the HSI classification task. A deep supervised learning classifier for HSI classification using CNNs (CNN classifier) [50] was chosen as our test model.

The framework for executing the CNN classifier on the HSI data is shown in Fig. 13. The original HSI data typically contain hundreds of bands. Thus, it is not suitable to be directly used as the input of a CNN, as too many input channels will

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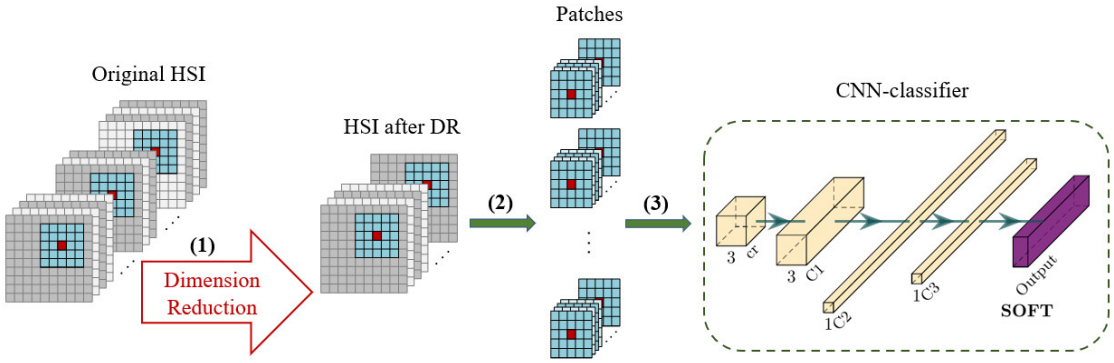


Fig. 13. Framework for executing the CNN classifier on the HSI data.

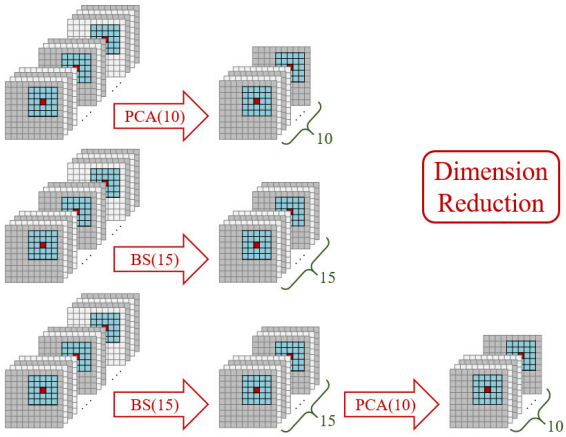


Fig. 14. Three ways to reduce the spectral dimension when generating data sets for the CNN classifier.

increase the training cost. Through dimensionality reduction, the spectral dimension of the original HSI was condensed into ten channels. New data sets were constructed by generating square patches of size 5×5 pixel centered at each pixel. The class label of each patch is denoted by the label of the centered pixels.

In our experiment, the tagged data sets were randomly split into three parts, i.e., training, validation, and testing data, with a ratio of $6 : 2 : 2$. Background pixels were not taken account into. The training data were fed into a CNN classifier to update the parameters of the CNN. The validation data were used to quantitatively validate the CNN in terms of classification accuracy. Finally, we chose the group of CNN parameters with the best validation result in the first 100 epochs as the best model. The testing data were sent to the best model, and the classification accuracy was the final test result.

While generating a data set for deep learning, DR on the original HSI data is usually implemented by principal components analysis (PCA) [50], [51]. To explore the application of BS in deep learning, we tried three ways to reduce the HSI dimension and compared their test accuracy. The three DR methods are introduced as follows and described in Fig. 14.

1) *PCA(10):* We reserved the first ten principal components (PCs) of the original-HSI to form the HSI-after-DR. Based on a previous publication [50], we chose the number of retained PCs to be 10 because the first ten PCs contained more than 99% of raw data information.
2) *BS(15):* We selected 15 bands from the hundreds of bands in the original-HSI and the 15 band images formed the HSI-after-DR. Only 15 bands were selected because we found that was sufficient for classification through the previous experiments of BS methods with conventional classifiers.
3) *BS(15) + PCA(10):* First, we selected 15 bands from the original HSI. Next, the first ten PCs of the HSI, which contained only 15 selected bands, were used to make up the HSI-after-DR data set.

2) Results and Analysis: Fig. 15 shows the classification accuracy curves of the training, validation, and testing data while using the three different DR methods on the four data sets. The BS method was our proposed SR-SSIM-w3. When using SR-SSIM-w3 and PCA in turn, the test accuracy was the highest on all four data sets, and the validation accuracy curves reached higher values than those when using only PCA on the Indian Pines, KSC, and Salinas data sets. By comparison, when only using BS(15), the results were not as good, except for the Salinas data set.

Furthermore, we compared the test accuracy when using the other BS methods described above, as shown in Tables II and III. For the Indian Pines data set, by using BS + PCA, the test results were better than those when using only PCA, except for the OPBS method. For the Salinas and KSC data sets, using BS alone or combining BS and PCA together could both improve the test results. For the Botswana data set, the test accuracy when using SR-SSIM-w3 + PCA reached 1.0, and the result of OCF + PCA was also better than that of PCA alone. In addition, BS alone and BS + PCA did not improve the results for the Botswana data set.

Through these experiments on the CNN classifier, we found that the BS method can be applied as a preprocessing step before PCA to reduce the dimension of HSI raw data for most data sets. Compared with using PCA alone, adding the BS process can significantly improve the results. This was because the original HSI data contained considerable

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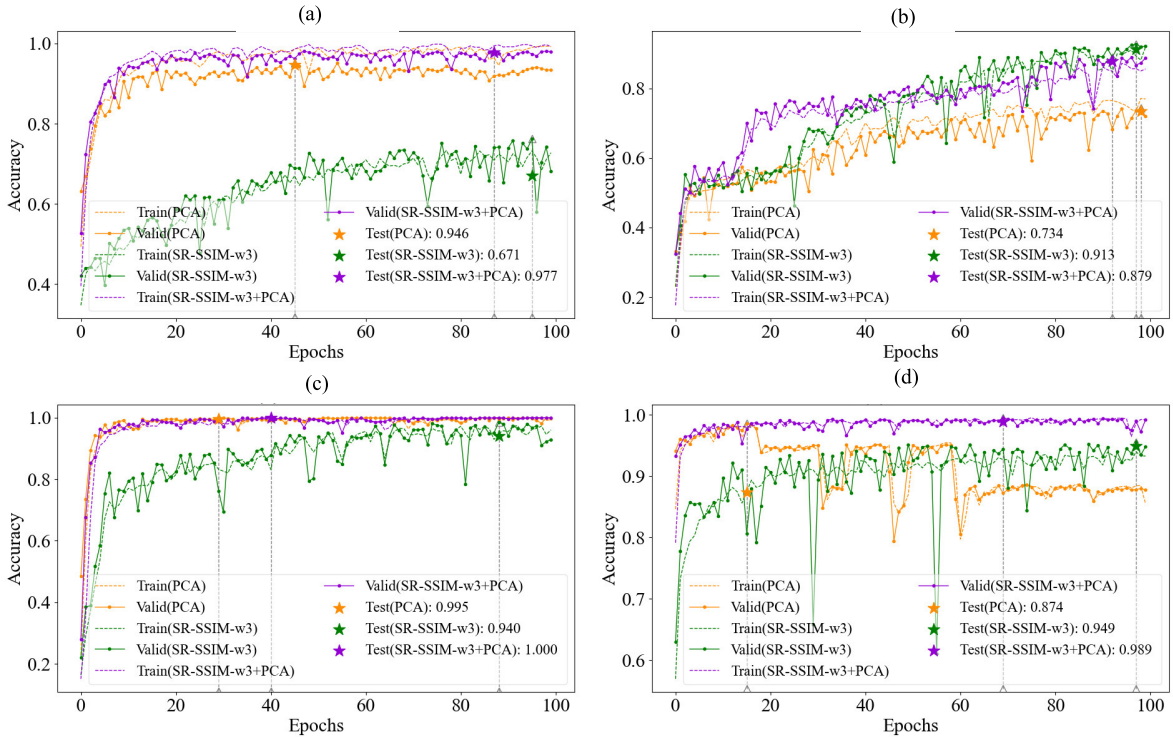


Fig. 15. Application of SSIM-FDPC on (a) Indian Pines, (b) KSC, (c) Botswana, and (d) Salinas data sets. The dashed and solid lines represent the training and validation accuracies, respectively. The five-pointed star shows the testing accuracy in the epoch with the largest validation accuracy. Orange, green, and violet represent the three different DR methods.

TABLE II

COMPARISON FOR PCA(10) AND BS(15): TEST RESULTS OF OUR PROPOSED AND OTHER COMPETITIVE BS METHODS WITH A CNN CLASSIFIER

DR-ways	PCA (10)	BS (15)						
	PCA	SR-SSIM	SR-SSIM-w3	EFDPC-SSIM	EFDPC	TRC-OC-FDPC	NC-OC-FDPC	OPBS
Indian pines	*0.946*	0.652	0.671	0.707	0.709	0.644	0.686	0.594
Salinas	*0.874*	**0.946**	**0.949**[*]	**0.879**	**0.930**	**0.941**	**0.898**	**0.878**
KSC	*0.734*	**0.922**	**0.913**	**0.858**	**0.920**	**0.867**	**0.861**	0.733
Botswana	*0.995*	0.972	0.940	0.951	0.815	0.971	0.971	0.898

[*] For each dataset, if the testing accuracy of BS(15) was larger than the testing accuracy of PCA(10), then the value is shown in bold. Meanwhile, the best result for each dataset is shown in red and bold.

TABLE III

COMPARISON FOR PCA(10) AND BS(15) + PCA(10): TEST RESULTS OF OUR PROPOSED AND OTHER COMPETITIVE BS METHODS WITH A CNN CLASSIFIER

DR-ways	PCA (10)	BS (15) + PCA (10)						
	PCA	SR-SSIM	SR-SSIM-w3	EFDPC-SSIM	EFDPC	TRC-OC-FDPC	NC-OC-FDPC	OPBS
Indian pines	*0.946*	**0.983**	**0.967**	**0.984**[*]	**0.949**	**0.983**	**0.982**	0.932
Salinas	*0.874*	**0.990**	**0.987**	**0.993**	**0.881**	**0.987**	**0.978**	**0.971**
KSC	*0.734*	**0.915**	**0.879**	**0.865**	**0.941**	**0.945**	**0.890**	0.554
Botswana	*0.995*	**1.0**	**1.0**	**0.998**	0.975	**0.997**	0.994	0.974

[*] For each dataset, if the testing accuracy of BS(15) + PCA(10) was larger than the testing accuracy of PCA(10), then the value is shown in bold. Meanwhile, the best result for each dataset is shown in red and bold.

redundant information in the spectral dimension, and BS processing could eliminate the interference of redundant bands when retaining some of the useful bands, avoiding overfitting. In addition, after BS, further DR by PCA can reduce the number of input channels of the CNN network, which can avoid excessive network parameters and training costs.

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V. Conclusion

In this article, we proposed an SR method for hyperspectral BS. The representativeness of each band was evaluated through the product of the average similarity and the significant dissimilarity. Unlike BS methods that only employ the spectral information of HSIs, our method combines the spatial information and spectral information by introducing the SSIM. Furthermore, we found from the experimental analysis that setting the window size in the SSIM index to three could balance the effectiveness and complexity of the hyperspectral BS problem. This similarity measurement method proved to be more effective than the Euclidean distance and can be used with other BS algorithms.

Through classification experiments, our proposed method was shown to be superior to EFDPC, TRC-OC-FDPC, NC-OC-FDPC, and OPBS. These competitive BS methods treat hyperspectral band images as points in high-dimensional Euclidean space. This perspective ignores the spatial information of each band image. For the ablation study, we replaced the Euclidean distance in EFDPC with $(1 - \text{SSIM})$ and obtained the EFDPC-SSIM algorithm. The results showed that EFDPC-SSIM could perform better than EFDPC but was still inferior to our proposed method. This not only demonstrated the advantage of SSIM over the Euclidean distance for hyperspectral BS but also presents the improvement effect of the SR on EFDPC.

Moreover, through experiments on a CNN classifier, we found that, when deep learning was used for HSI classification, proper preprocessing by the BS method could effectively eliminate redundant information and avoid overfitting.

Furthermore, this study showed that the measurement of band similarity is very important for the efficiency of BS algorithms. In the future, we will continue to explore more suitable band similarity metrics for HSIs. Furthermore, we will continue to consider how to combine the spatial information and spectral information for efficient BS.

Acknowledgment

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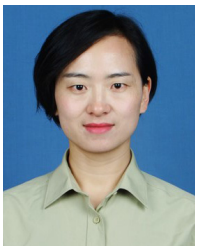
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